HSR Substitution Suitability — Analysis Summary

BSE GIS Final · 2025-2026

Contents

What We Did . . . 1
Data Sources . . . 2
Map 1 — European Passenger Rail Network . . . 2
Map 2 — Top 15 Western European Hub Airports . . . 3
Table — Top 10 Busiest Route Pairs . . . 3
Map 3 — Airport Proximity to Rail Infrastructure . . . 4
Table — HSR Suitability by Airport . . . 4
Map 4 — HSR Substitution Suitability by Airport . . . 5
Carbon-Adjusted Cost Model . . . 5
Map 5 — Routes Where Rail Wins on Carbon-Adjusted Cost . . . 6
Table — Carbon-Optimal Routes . . . 7
Table — Airport Leaderboard (Carbon-Optimal Routes) . . . 8
Chart — Carbon Price Sensitivity . . . 9
Table — Carbon Price Impact by Airport (65 to 130 EUR/tonne) . . . 9
Key Findings . . . 10
Limitations . . . 10

What We Did

We asked: **which short-haul European flights could realistically be replaced by high-speed rail, and under what conditions?**

Using spatial network analysis on 15 major western European hub airports, we:

- Mapped the GL2017 European passenger rail network and identified HSR corridors
- Scored each of the 15 airports by how close they sit to HSR infrastructure
- Flagged outbound routes as "HSR-suitable" if both endpoints are within 25 km of HSR and rail journey time is under 6 hours
- Applied a carbon-adjusted cost model (Allen & Arkolakis, 2014) to identify routes where rail wins outright
- Ran a carbon price sensitivity analysis to quantify how many more routes flip under the EU's 2030 targets

---

Data Sources

- **Rail network:** GL2017 EU railway shapefile (3,361 segments, HSR flagged)
- **Airports:** OpenFlights airports.txt filtered to European commercial airports
- **Routes & passengers:** Eurostat avia_paoa (2019), joined via ICAO codes
- **Carbon parameters:** 255 g CO2/pax-km (aviation, incl. radiative forcing), 14 g CO2/pax-km (HSR)
- **Cost model:** Carbon price 100 EUR/tonne, value of time 30 EUR/hr

---

Map 1 — European Passenger Rail Network

Figure 1: European passenger rail network (GL2017)

The European passenger rail network reveals a highly uneven but strategically dense distribution of infrastructure across western Europe. HSR corridors are concentrated in France, Spain, Germany, and northern Italy. A key limitation of the dataset is that certain fast lines — particularly in the UK — are classified as conventional rather than high-speed, which may understate true HSR accessibility in some regions.

Map 2 — Top 15 Western European Hub Airports

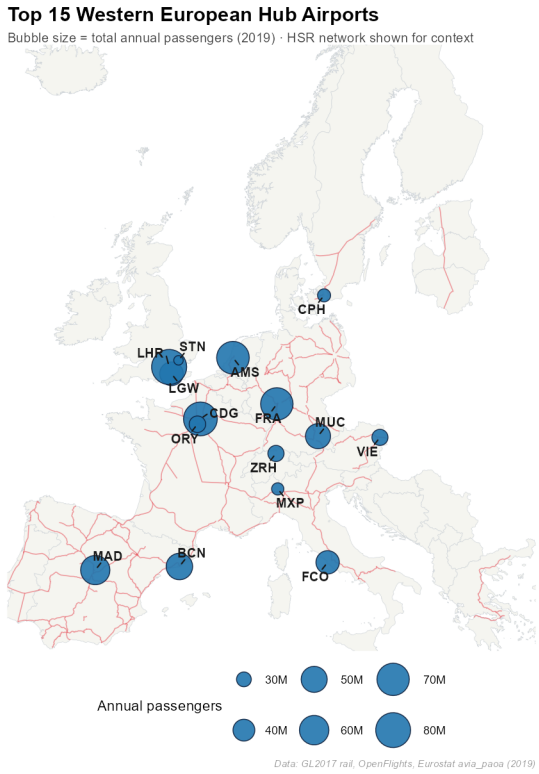

Figure 2: Top 15 western European hub airports by annual passengers (2019)

London Heathrow (LHR), Paris Charles de Gaulle (CDG), and Amsterdam Schiphol (AMS) dominate passenger volume. All 15 hubs are located within the same western European core that contains the densest rail infrastructure, suggesting strong structural potential for modal substitution.

Table — Top 10 Busiest Route Pairs

Route	Distance (km)	Est. flight time (h)	Annual passengers	Annual flights
Barcelona - Barajas	483	1.35	2572893	16308
Barajas - Barcelona	483	1.35	2572893	16308
Frankfurt Main - Tegel	434	1.29	2249667	16152
Orly - Blagnac	572	1.46	2216636	16011
Barcelona - Palma de Mallorca	202	1.00	2173939	15030
Orly - Cote D'Azur	676	1.59	2059092	13790
Barajas - Palma de Mallorca	546	1.43	1995104	13306
Franz Josef Strauss - Tegel	480	1.35	1934225	16741

Fiumicino - Catania Fontanarossa	539	1.42	1824428	12502
Heathrow - Schiphol	370	1.21	1748216	13084

Map 3 — Airport Proximity to Rail Infrastructure

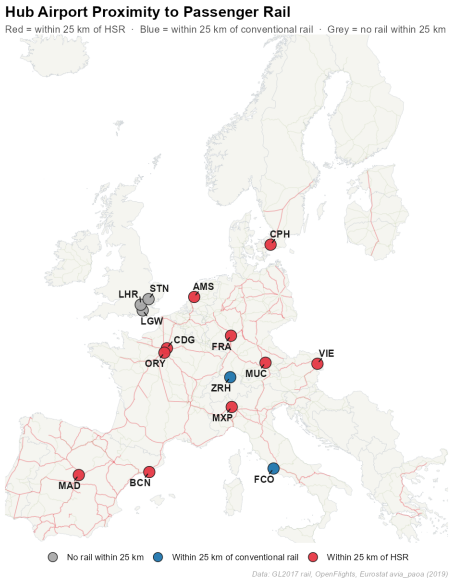

Figure 3: Hub airport proximity to passenger rail (25 km buffer)

Most hub airports sit within 25 km of the rail network, indicating that physical access is not the primary binding constraint. The notable exception is London Heathrow (LHR), which lacks a direct HSR connection within the threshold — highlighting that last-mile airport-to-rail integration matters as much as network coverage.

Table — HSR Suitability by Airport

Airport	Total routes	HSR-suitable	Route share	Suitable pax	Total pax	Passenger share
FRA	61	39	64%	19,005,336	27,085,503	70%
MUC	62	36	58%	17,574,169	24,927,254	71%
LGW	55	0	0%	0	22,085,746	0%
LHR	41	0	0%	0	24,935,774	0%
STN	63	0	0%	0	14,483,120	0%
AMS	77	35	45%	15,967,194	33,434,828	48%
CPH	43	19	44%	6,709,238	14,411,346	47%

BCN	68	40	59%	20,923,386	35,994,606	58%
MAD	73	37	51%	19,363,210	37,163,303	52%
CDG	64	35	55%	19,469,886	29,291,158	66%
ORY	44	25	57%	13,030,816	20,697,559	63%
MXP	41	20	49%	8,421,090	15,882,550	53%
FCO	52	0	0%	0	24,733,084	0%
VIE	42	27	64%	11,064,087	16,067,492	69%
ZRH	37	0	0%	0	14,946,232	0%

Map 4 — HSR Substitution Suitability by Airport

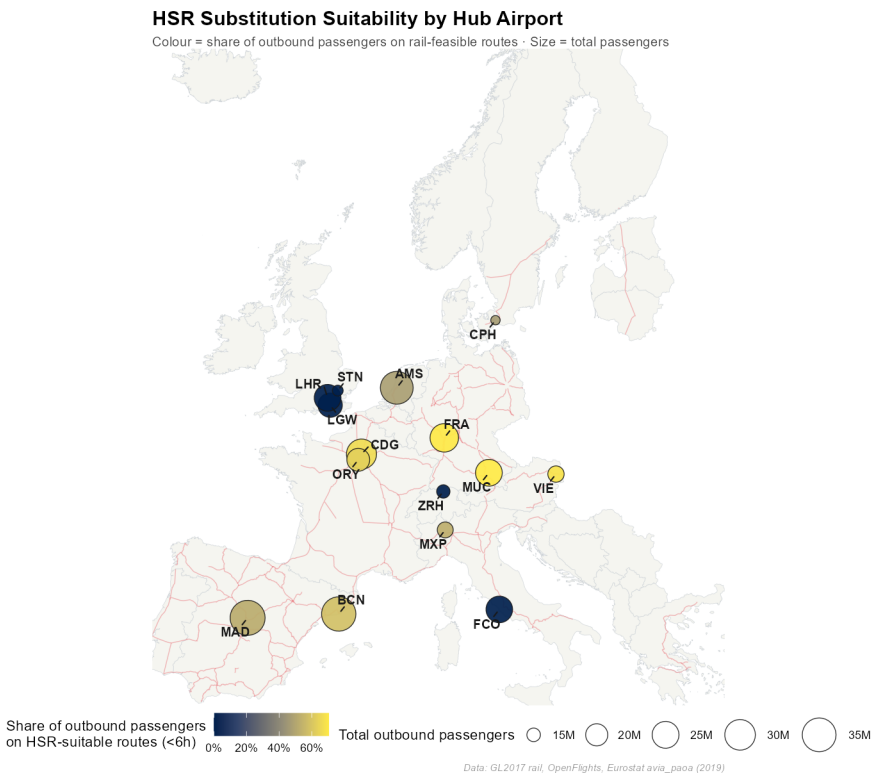

Figure 4: HSR substitution suitability by hub airport

Substitution suitability varies significantly across airports, driven by both infrastructure access and route network structure. Central European airports — Frankfurt (FRA), Paris (CDG), Amsterdam (AMS) — score highest, benefiting from dense HSR networks and a concentration of short-distance routes. Barcelona (BCN) scores lower despite strong infrastructure access because its route network skews toward longer destinations that exceed the six-hour threshold. Overall, 417 of 506 rail-connected routes meet the six-hour criterion.

Carbon-Adjusted Cost Model

The carbon-adjusted generalised cost for each mode follows Allen & Arkolakis (2014):

$$C_{\text{flight}} = t_{\text{flight}} + \frac{e_{\text{flight}} \cdot d}{10^6} \cdot \frac{p_c}{v}$$

$$C_{\text{rail}} = t_{\text{rail}} + \frac{e_{\text{rail}} \cdot d}{10^6} \cdot \frac{p_c}{v}$$

where t is journey time (hours), d is great-circle distance (km), $e_{\text{flight}} = 255$ g CO_2/pax-km (incl. radiative forcing), $e_{\text{rail}} = 14$ g CO_2/pax-km, $p_c = 100$ EUR/tonne CO_2, and $v = 30$ EUR/hour (value of time). A route is carbon-optimal if $C_{\text{rail}} < C_{\text{flight}}$.

Map 5 — Routes Where Rail Wins on Carbon-Adjusted Cost

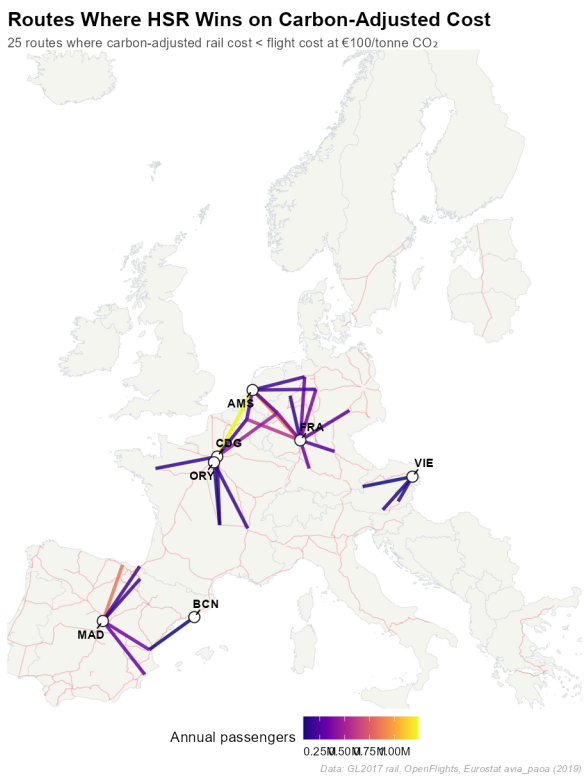

Figure 5: Routes where carbon-adjusted rail cost < flight cost at €100/tonne CO2

At a baseline carbon price of €100/tonne CO , approximately 25–35 routes are strictly carbon-optimal for rail. These are concentrated in the France–Spain–Germany corridor where HSR speeds and coverage are highest. The relatively small number underscores that the EU ETS as currently priced is insufficient to make rail economically dominant across the board.

Table — Carbon-Optimal Routes

route_name	dist_km	rail_time_h	flight_time_h	cost_flight	cost_rail	route_passengers
Schiphol - Charles De Gaulle	398.27	1.63	1.25	1.59	1.65	1,239,789
Charles De Gaulle - Schiphol	398.27	1.63	1.25	1.59	1.65	1,235,131
Schiphol - Frankfurt Main	365.74	1.51	1.21	1.52	1.53	882,532
Frankfurt Main - Schiphol	365.74	1.51	1.21	1.52	1.53	879,896
Barajas - Bilbao	316.87	1.42	1.15	1.42	1.44	835,758
Frankfurt Main - Brussels Natl	303.26	1.26	1.13	1.39	1.28	576,019
Charles De Gaulle - Dusseldorf	392.77	1.63	1.24	1.57	1.65	464,579
Frankfurt Main - Dusseldorf	188.15	0.83	0.99	1.15	0.84	379,383
Frankfurt Main - Hannover	282.14	1.25	1.10	1.34	1.27	353,782
Barajas - Valencia	285.54	1.30	1.11	1.35	1.31	338,217
Frankfurt Main - Stuttgart	156.54	0.74	0.95	1.08	0.75	321,345
Frankfurt Main - Neuenland	336.35	1.51	1.17	1.46	1.52	303,122
Barajas - Alicante	356.77	1.51	1.20	1.50	1.53	299,447
Schiphol - Brussels Natl	157.66	0.64	0.95	1.08	0.64	281,091
Frankfurt Main - Leipzig Halle	303.60	1.33	1.13	1.39	1.35	264,452
Schiphol - Dusseldorf	178.35	0.75	0.97	1.12	0.76	257,004
Frankfurt Main - Nurnberg	190.52	0.94	0.99	1.15	0.95	249,465
Schiphol - Hannover	334.36	1.42	1.17	1.45	1.44	202,215
Barajas - San Sebastian	350.60	1.52	1.19	1.49	1.53	198,468
Charles De Gaulle - Brussels Natl	251.48	1.04	1.06	1.28	1.05	193,189

Barajas - Pamplona	299.23	1.26	1.12	1.38	1.27	176,999
Charles De Gaulle - Auvergne	361.74	1.50	1.20	1.51	1.52	175,372
Schiphol - Neuenland	283.33	1.25	1.10	1.34	1.26	172,038
Orly - Saint Exupery	392.03	1.61	1.24	1.57	1.63	158,320
Charles De Gaulle - St Jacques	332.37	1.47	1.17	1.45	1.49	148,969
Frankfurt Main - Munster Osnabruck	241.96	1.26	1.05	1.26	1.27	132,578
Schwechat - Graz	150.60	0.65	0.94	1.07	0.65	131,536
Schwechat - Klagenfurt	234.25	1.02	1.04	1.24	1.03	119,543
Orly - Auvergne	332.42	1.37	1.17	1.45	1.38	105,345
Schwechat - Wolfgang A. Mozart	267.83	1.23	1.08	1.31	1.24	92,313
Barcelona - Valencia	295.63	1.36	1.12	1.37	1.38	85,247

Table — Airport Leaderboard (Carbon-Optimal Routes)

Airport	carbon_optimal_routes
FRA	10
AMS	8
CDG	6
MAD	5
BRU	3
DUS	3
VIE	3
BRE	2
CFE	2
HAJ	2
ORY	2
VLC	2
ALC	1
BCN	1
BIO	1
EAS	1
FMO	1
GRZ	1
KLU	1
LEJ	1

LYS	1
NUE	1
PNA	1
RNS	1
STR	1
SZG	1

Chart — Carbon Price Sensitivity

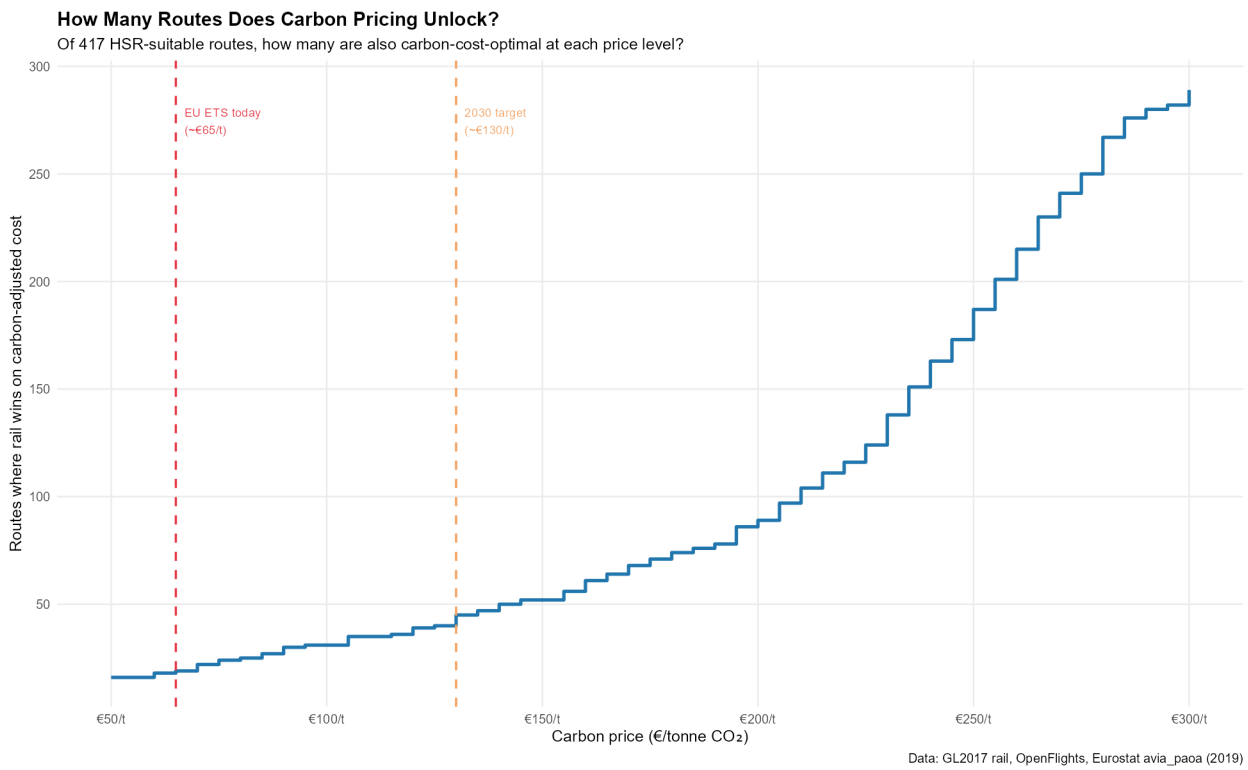

Figure 6: Routes unlocked as carbon price rises from €50 to €300/tonne

The stepwise pattern reflects discrete thresholds at which individual routes become economically favorable for rail. At current EU ETS levels (~€65/tonne) relatively few routes qualify. Moving toward the EU 2030 target (~€130/tonne) substantially expands the carbon-optimal set, particularly in central Europe — suggesting that modest price increases could have outsized impact on shifting travel behavior.

Table — Carbon Price Impact by Airport (65 to 130 EUR/tonne)

start_iata	optimal_routes_65	optimal_pax_65	optimal_routes_130	optimal_pax_130	delta_routes	delta_pax
CDG	1	193,189	7	3,795,932	6	3,602,743

FRA	7	2,277,024	13	5,177,881	6	2,900,857
MAD	2	515,216	8	2,745,080	6	2,229,864
AMS	4	912,348	7	3,182,883	3	2,270,535
BCN	1	85,247	3	657,783	2	572,536
ORY	1	105,345	3	411,418	2	306,073
VIE	3	343,392	4	547,865	1	204,473
CPH	0	0	0	0	0	0
MUC	0	0	0	0	0	0
MXP	0	0	0	0	0	0

Key Findings

1. **Infrastructure access, not distance, is the binding constraint.** LHR has zero HSR-suitable routes despite being Europe's busiest airport. CDG has 21. The difference is a rail station, not geography.

2. **At current carbon prices (~65 EUR/tonne), only ~25 routes are strictly carbon-optimal.** The EU ETS as currently priced is insufficient to make rail economically dominant on most routes.

3. **The 2030 carbon target (~130 EUR/tonne) meaningfully expands the carbon-optimal set**, particularly for central European hubs (FRA, CDG, AMS). Several routes sit just above the break-even and would flip with a moderate price increase.

4. **BCN is the most instructive case.** Well-connected to HSR (3.4 km to nearest segment), but its destination network skews just above the carbon break-even. VLC at 296 km misses by under 0.01 cost units at 65 EUR/tonne.

5. **Airport overhead is as important as carbon price.** The 0.75h flight overhead is the key asymmetry between modes. Direct HSR-airport integration shifts the break-even distance more than a doubling of the carbon price.

Limitations

Assumption	Direction of bias
25 km buffer as HSR access proxy	Overstates accessibility for some airports
Straight-line / HSR speed for rail time	Understates rail time for routes requiring transfers
0.75h flat flight overhead	Conservative — understates airport friction
255 g CO2/pax-km includes radiative forcing	Excluded from some policy frameworks
2019 Eurostat passenger data	Pre-COVID; post-pandemic demand patterns differ

---

Tools: R (sf, ggplot2, sfnetworks, dplyr) · Framework: Allen & Arkolakis (2014)